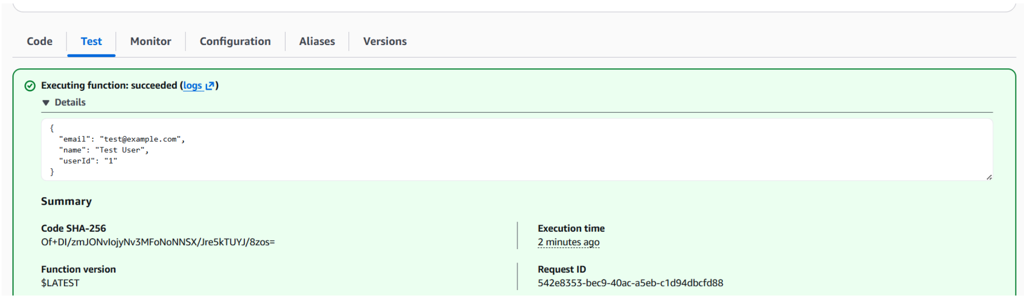
FETCH DATA WITH AWS LAMBDA

By Mohd Mudassir Arfath | July 2026

Fetch Data with AWS Lambda



Introducing Today's Project!

In this project, we will learn how to use an AWS Lambda function to retrieve data from a DynamoDB table. We will create a Lambda function, connect it to DynamoDB, and write code to fetch the stored data. Finally, we will test the function and check the retrieved results.

Tools and concepts

The key services and concepts I learned in this project were AWS Lambda, Amazon DynamoDB, IAM roles and permissions, AWS SDK, serverless computing, and testing Lambda functions. I learned how to connect Lambda with DynamoDB and retrieve data securely using the correct permissions.

Project reflection

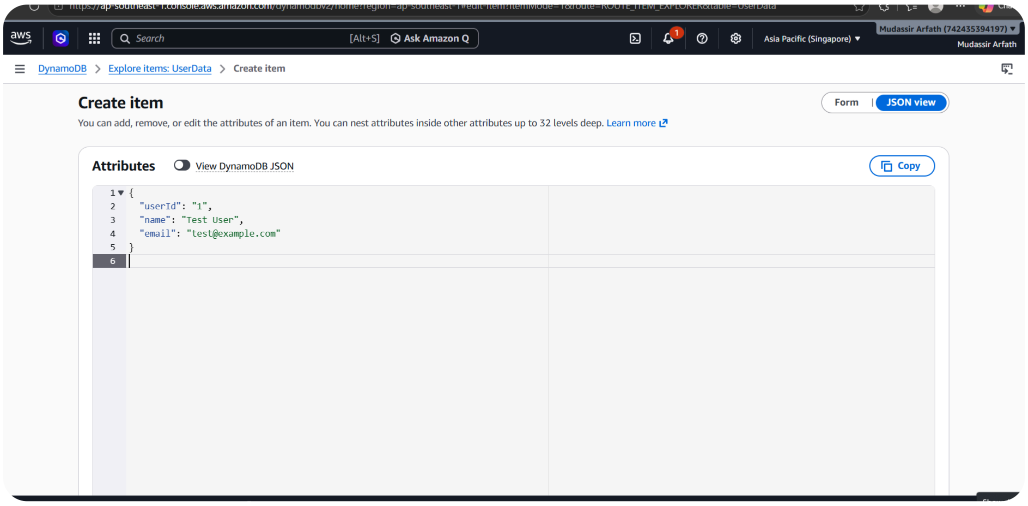
It took me approximately 1–2 hours to complete this project, including setting up DynamoDB, creating and configuring the Lambda function, granting permissions, and testing the function.

Thank you! I enjoyed completing this project and learning how to use AWS Lambda with DynamoDB to retrieve data. It was a great hands-on experience with serverless computing and AWS IAM permissions

Project Setup

I logged in to the AWS Management Console as an IAM Admin user, selected the AWS region closest to me, and opened the DynamoDB console to create and manage the DynamoDB table.

I added a sample user item to my UserData DynamoDB table, which will be used to test the Lambda function and retrieve the stored user data.



AWS Lambda

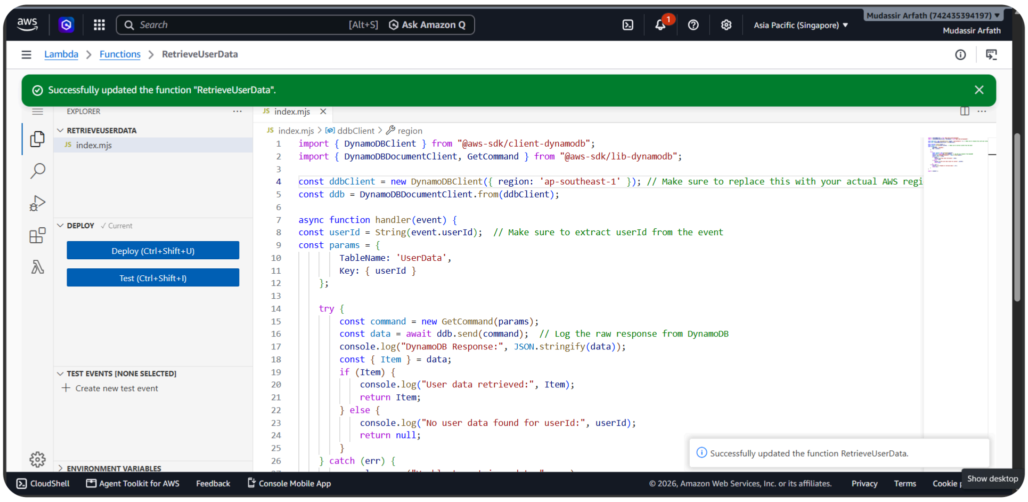
AWS Lambda is a serverless computing service that lets you run code without managing servers. It automatically runs your code when triggered and you only pay for the compute time you use.

AWS Lambda Function

The Lambda function's execution role gives the function permission to access AWS services and resources it needs. In this project, it allows the Lambda function to read data from the DynamoDB table.

The code block connects to the DynamoDB table and retrieves the stored user data. It then returns the retrieved data as the output of the AWS Lambda function.

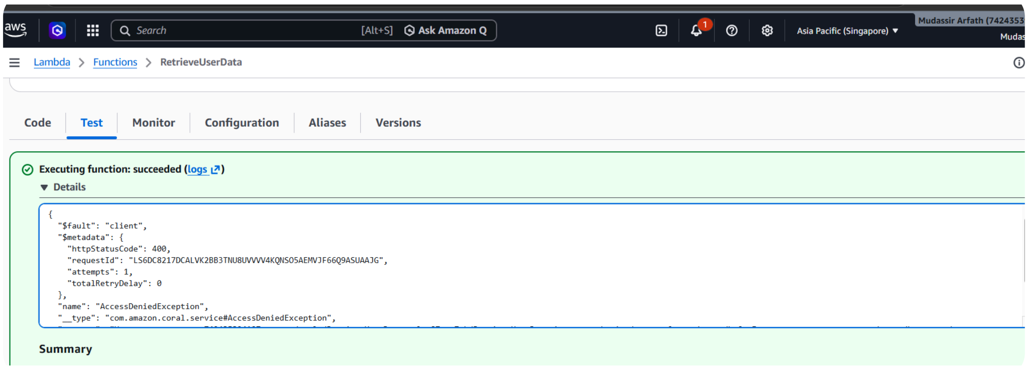
AWS SDK is a collection of tools and libraries that allows developers to interact with AWS services, such as DynamoDB, S3, and Lambda, using programming languages like Python or JavaScript.



Function Testing

I tested my Lambda function by creating a test event in the AWS Lambda console and running the function. I then checked the test results to confirm that it successfully retrieved the user data from the DynamoDB table.

I ran into an error because the Lambda function did not have the correct permissions to access the DynamoDB table. The execution role needed the required DynamoDB permissions for the function to retrieve the data successfully.

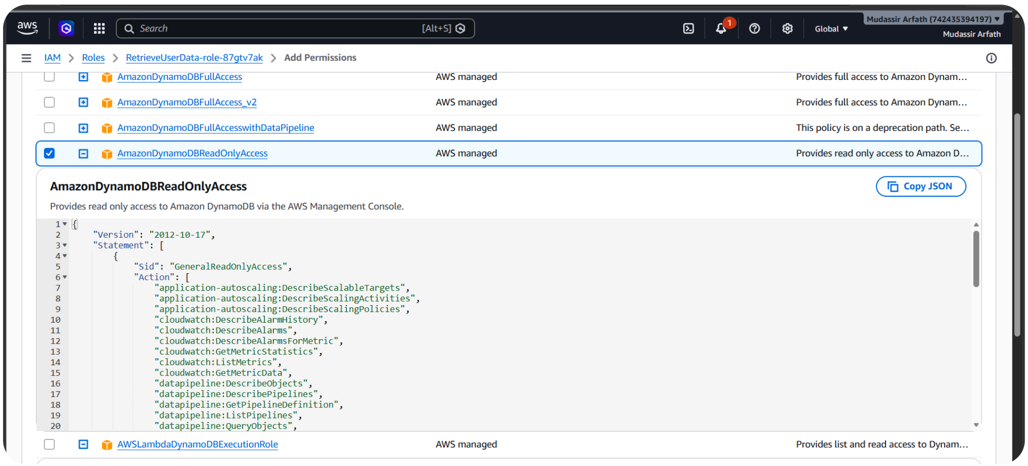


Function Permissions

I am resolving the Access Denied error by attaching the AmazonDynamoDBReadOnlyAccess policy to the Lambda function's execution role. This gives the Lambda function permission to read data from the DynamoDB table.

I wouldn't select those two policies because they provide more permissions than the Lambda function needs. Our function only needs to read data from DynamoDB, so AmazonDynamoDBReadOnlyAccess follows the principle of least privilege and is the safer choice.

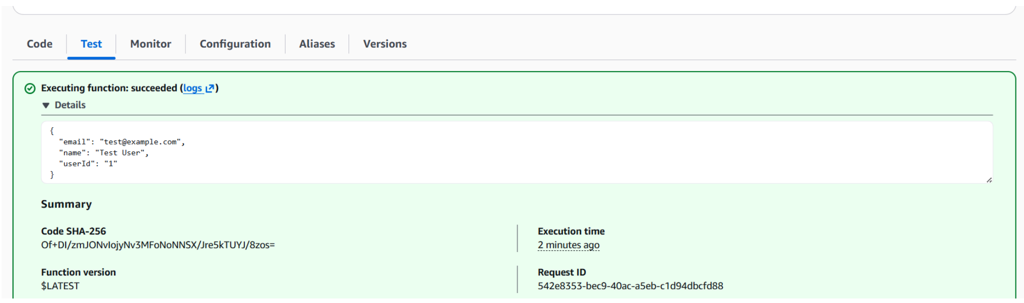
I picked AmazonDynamoDBReadOnlyAccess because the Lambda function only needs to read and retrieve data from the DynamoDB table. It does not need permission to add, update, or delete data.



Final Testing and Reflection

I saw a successful result because I granted the Lambda function permission to read from DynamoDB by attaching the AmazonDynamoDBReadOnlyAccess policy to its execution role. This allowed the function to successfully retrieve the user data from the DynamoDB table.

I can use what I learned to build serverless applications where AWS Lambda retrieves and processes data from DynamoDB. I can also use these skills to create APIs, user management systems, and cloud-based applications using AWS services...



Enahancing Security

We are replacing the Lambda function's permission policy because the previous AmazonDynamoDBReadOnlyAccess policy gives broad read access to DynamoDB resources. We want to follow the principle of least privilege by giving the Lambda function only the specific permissions it needs to access our UserData table, making the database more secure.

I picked the inline policy setup method because it allows me to give the Lambda function only the specific permissions it needs to access the DynamoDB UserData table. This follows the principle of least privilege and improves security.

In this step, we are testing the Lambda function again after tightening its permissions. We run the test to confirm that the function still has the required access to the DynamoDB UserData table and can successfully retrieve the table items....